

☒ 40 MCQs: Regular Expressions, Scripting, and Functional Programming

☒ Easy Level (10 Questions)

Q1. Which module is used for regular expressions in Python?

- A. regex
- B. re
- C. pyregex
- D. express

Answer: B

Explanation: Python's built-in module for regex is re.

Q2. What does re.findall() return?

- A. The first match only
- B. A list of all matches
- C. Boolean True/False
- D. Replaces matches

Answer: B

Explanation: Returns all non-overlapping matches.

Q3. What is the symbol for a digit in regex?

- A. \w
- B. \d
- C. .
- D. \s

Answer: B

Explanation: \d matches any digit.

Q4. What is the purpose of the if `__name__ == "__main__":` statement?

- A. To run code only when script is executed directly
- B. To import modules
- C. To declare a main function
- D. To start a loop

Answer: A

Explanation: It checks if script is run directly.

Q5. How do you run a Python script named test.py from the command line?

- A. run test.py
- B. python test.py
- C. exec test.py
- D. start test.py

Answer: B

Explanation: python filename.py runs the script.

Q6. What keyword defines an anonymous function in Python?

- A. def
- B. lambda
- C. function
- D. anon

Answer: B

Explanation: lambda creates anonymous functions.

Q7. What does map() do?

- A. Applies a function to every item in an iterable
- B. Filters items based on condition
- C. Reduces iterable to a single value
- D. Sorts items

Answer: A

Explanation: map() applies function on all elements.

Q8. Which function is used to accumulate values from an iterable?

- A. map()
- B. filter()
- C. reduce()

D. accumulate()

Answer: C

Explanation: reduce() applies a rolling computation.

Q9. What does re.sub() do?

A. Searches for a pattern

B. Finds all matches

C. Replaces matched text

D. Splits strings

Answer: C

Explanation: Substitutes matched patterns.

Q10. Which function is used to filter elements based on a condition?

A. map()

B. filter()

C. reduce()

D. select()

Answer: B

Explanation: filter() selects elements that satisfy a condition.

☒ **Medium Level (20 Questions)**

Q11. What will this regex pattern match? `r"^\d{3}-\d{2}-\d{4}$"`

- A. Any number
- B. Social Security Number format like 123-45-6789
- C. Email address
- D. Date

Answer: B

Explanation: Matches exactly 3 digits, dash, 2 digits, dash, 4 digits.

Q12. What does this code output?

```
python
```

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```
import re
text = "abc123xyz"
result = re.search(r"\d+", text)
print(result.group())
```

- A. abc
- B. 123

- C. xyz
- D. None

Answer: B

Explanation: \d+ matches one or more digits.

Q13. What will `lambda x: x % 2 == 0` return when used with `filter()` on `[1, 2, 3, 4]`?

- A. `[1, 3]`
- B. `[2, 4]`
- C. `[1, 2, 3, 4]`
- D. `[]`

Answer: B

Explanation: Filters even numbers.

Q14. How do you pass command-line arguments in Python scripts?

- A. Using `sys.argv`
- B. Using `argparse` module
- C. Both A and B
- D. Using `input()` function

Answer: C

Explanation: Both provide ways to handle arguments.

Q15. Which of these is a correct way to write a function that returns the square of a number using `lambda`?

- A. `square = lambda x: x**2`
- B. `def square(x): return x**2`
- C. Both A and B
- D. None

Answer: C

Explanation: Both are valid.

Q16. What will this code do?

python

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```
from functools import reduce print(reduce(lambda x, y: x + y, [1, 2, 3, 4]))
```

- A. 10
- B. 24
- C. [1, 2, 3, 4]
- D. Error

Answer: A

Explanation: Sums all list elements.

Q17. What happens if `re.match()` does not find a match?

- A. Returns None
- B. Raises exception
- C. Returns empty string
- D. Returns False

Answer: A

Explanation: None if no match at start.

Q18. Which statement about functional programming is false?

- A. Functions are first-class objects
- B. Functions cannot be assigned to variables
- C. Supports anonymous functions
- D. Encourages immutability

Answer: B

Explanation: Functions can be assigned to variables.

Q19. How can you handle errors in Python scripting?

- A. Using try-except blocks
- B. Using if-else only
- C. Using catch keyword
- D. Not possible

Answer: A

Explanation: try-except is used for error handling.

Q20. What is the output?

python

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```
print(list(map(str, [1, 2, 3])))
```

- A. ['1', '2', '3']
- B. [1, 2, 3]
- C. ['str', 'str', 'str']
- D. Error

Answer: A

Explanation: Converts each number to string.

🚩 **Hard Level (10 Questions)**

Q21. What will this regex do?

python

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```
re.findall(r"(\w+)@(\w+)\.(\w+)", "test@example.com")
```

- A. Matches email and returns groups: [('test', 'example', 'com')]
- B. Returns entire string
- C. Error
- D. Returns None

Answer: A

Explanation: Captures username, domain, and TLD groups.

Q22. What is a key benefit of functional programming?

- A. Easier mutable state management
- B. Code is more predictable and testable
- C. Uses loops more effectively
- D. Requires less memory

Answer: B

Explanation: Functional programming avoids side effects.

Q23. Which built-in function can convert a string to an integer?

- A. int()
- B. str()
- C. float()
- D. ord()

Answer: A

Explanation: int() converts strings to int.

Q24. What does this snippet print?

python

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```
import re print(re.sub(r"\d", "#", "abc123"))
```

A. abc###

B. abc123

C. abc

D. Error

Answer: A

Explanation: Replaces digits with #.

Q25. What is the difference between `re.match()` and `re.search()`?

A. `match()` looks at start only, `search()` anywhere

B. No difference

C. `search()` looks at start only

D. `match()` returns list

Answer: A

Explanation: `match()` requires pattern at start.

Q26. What will this print?

```
python
```

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```
def apply_func(f, value): return f(value) print(apply_func(lambda x: x*3, 5))
```

- A. 15
- B. 5
- C. lambda x: x*3
- D. Error

Answer: A

Explanation: Function applied to value.

Q27. What does `functools.reduce()` require as first argument?

- A. A function with two arguments
- B. A list
- C. A lambda function only
- D. A number

Answer: A

Explanation: Reducer function takes two args.

Q28. How to import only `reduce` from `functools`?

- A. `from functools import reduce`
- B. `import functools.reduce`
- C. `import reduce from functools`
- D. `functools.reduce import`

Answer: A

Explanation: Correct import syntax.

Q29. Which of the following matches any whitespace character?

- A. \w
- B. \s
- C. .
- D. \d

Answer: B

Explanation: \s matches whitespace.

Q30. Which regex pattern matches a string that starts and ends with digits?

- A. ^\d.*\d\$
- B. \d\$
- C. ^\d\$
- D. .* \d

Answer: A

Explanation: ^ and \$ anchor start and end.